

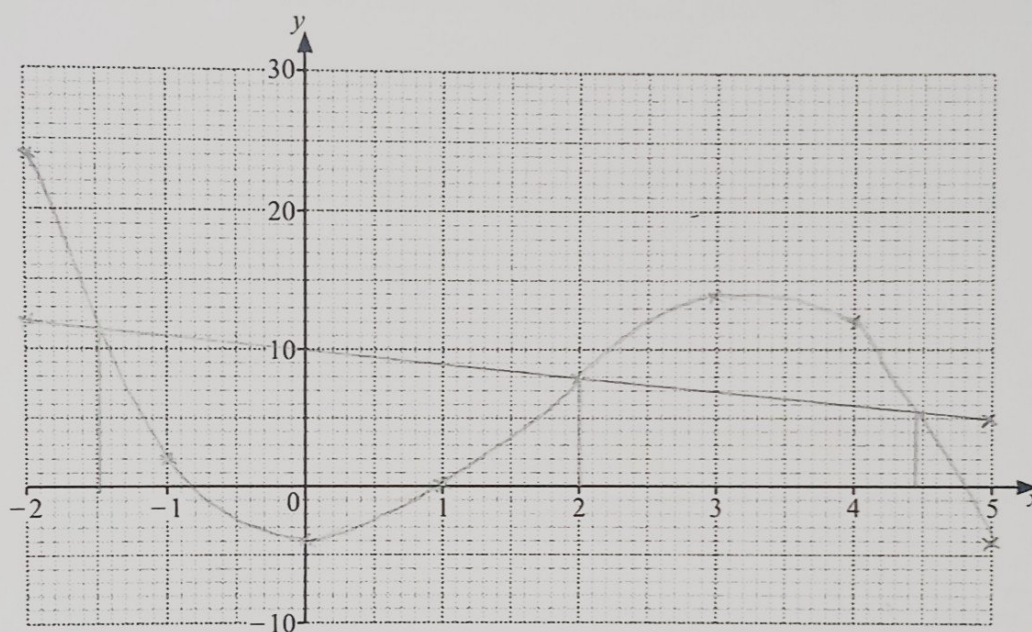
14 The table shows some values for $y = 5x^2 - x^3 - 4$.

x	-2	-1	0	1	2	3	4	5
y	24	2	-4	0	8	14	20	-4

(a) Complete the table.

[3]

(b) On the grid, draw the graph of $y = 5x^2 - x^3 - 4$ for $-2 \leq x \leq 5$.



[4]

(c) By drawing a suitable straight line on the grid, solve the equation $x^3 - 5x^2 - x + 14 = 0$.

$$= 5x^2 - x^3 - 4 + (x^3 - 5x^2 - x + 14)$$

$$= 5x^2 - 5x^2 - x^3 + x^3 - 4 + 14 - x$$

$$y = 10 - x$$

$$= -x + 10$$

$$x = -1.5 \text{ or } x = 2 \text{ or } x = 4.5 \quad [4]$$



- 15 The height of each of 140 basketball players is recorded.
The table shows the results.

	$160 < h \leq 180$	$180 < h \leq 185$	$185 < h \leq 190$	$190 < h \leq 200$	$200 < h \leq 210$
Height (h cm)					
Frequency	7	12	31	70	20

- (a) Calculate an estimate of the mean height.

M	170	182.5	187.5	195	205
f	7	12	31	70	20
$h \times f$	1190	2190	5812.5	13790	4100

$$= 140$$

$$= 27082.5$$

$$\text{mean} = \frac{\text{Total } hf}{\text{Total } f} = \frac{27082.5}{140}$$

$$= 193.45$$

$$\dots\dots\dots 193.45 \dots\dots\dots \text{cm} [4]$$

- (b) Two of the players are chosen at random.

Find the probability that both players have a height greater than 190 cm and no more than 200 cm.

$$= \frac{70}{140} \times \frac{69}{139}$$

$$= \frac{69}{278}$$

$$\dots\dots\dots \frac{69}{278} \dots\dots\dots [2]$$

$$f_d = \frac{7}{20}$$

$$= 0.35$$

$$f_d = \frac{12}{5}$$

$$= 2.4$$

$$f_d = \frac{31}{5}$$

$$= 6.2$$

$$f_d = \frac{70}{20}$$

$$= 3.5$$

$$f_d = \frac{20}{20}$$

$$= 1$$

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